MIL-S-14992A (AR)
AMENDMENT 9
24 OCTOBER 1979
SUPERSEDING
AMENDMENT 8
13 SEPTEMBER 1978

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M118
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-S-14992A (AR) dated 8 August 1974.

PAGE 2

Paragraph 3.5.1, change to read as follows:

"3.5.1 The S & A Mechanism assembly (9318159) shall exhibit the following resistance measurements when tested as specified in 4.4.2.1."

PAGE 3

Paragraph 3.5.1.1, change as follows:

Was: "threaded, electrical contact (9233189)"

Now: "contact, electrical, No 3 (9318162)"

Paragraph 3.5.1.2, change as follows:

Was: "electrical contact - No 2 in the base (9220897)

Now: "contact, electrical, stud (9318166)"

Was: "threaded electrical contact (9233189)"

Now: "contact, electrical, No 3 (9318162)"

FSC: 1336

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Paragraph 3.5.2.1, change as follows:

Was: "contact receptacle (9233186)"

Now: "contact, electrical, No 3 (9318162)"

Paragraph 3.5.2.2, change as follows:

Was: "contact receptacle (9233186) of the connector"

Now: "contact, electrical, No 3 (9318162)"

Was: "electrical, receptacle connector plug (9220901-1)"

Now: "contact, electrical, stud (9318166)"

Paragraph 3.6.1, change as follows:

S & A mechanism drawing number

Was: "9220959"

Now: "9318159"

Paragraph 3.7.1, change as follows:

S & A mechanism drawing number

Was: "9220959"
Now: "9318159"

Paragraph 3.7.2.1, in last line make change as follows:

Was: 537.0 milliseconds
Now: 560.0 milliseconds

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Paragraph 3.7.2.2, in last line make change as follows:

Was: 655.0 milliseconds
Now: 700.0 milliseconds

PAGE 5

Paragraph 3.9.1 Change referenced paragraph from "4.4.6" to "4.4.6.1".

Paragraph 3.9.2 Change to read as follows:

"3.9.2 S&A BASE ASSEMBLY (SEE DWG 9318160). There shall be no Breakdown of the insulation between the access cover assembly and electrical contact #3 (Internal S&A ground) as specified in 4.4.6.2".

PAGE 6

Add Para 3.18 as follows:

"3.18 Data requirements.-The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.4.22 and 4.4.23."

Add Para 3.19 as follows:

"3.19 Open Circuit Resistance Between Internal S & A Ground and the Nose-Crush (Missile) Ground.-With the S & A Device in the armed condition, the open circuit electrical resistance between the electrical contact, No 3 and the electrical receptacle connector plug (9220901-2) shall be no less than 100 megohms when tested as specified in 4.4.2.2.1".

PAGE 9

Paragraph 4.3.1, change as follows:

Was:
"a. Piston actuators from not more than one inspection lot number from one manufacturer.

b. Electric detonators from not more than one inspection lot number from one manufacturer."

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Now:

"a. Piston actuators from not more than two inspection lot numbers from one manufacturer.

b. Electric detonators from not more than one inspection lot number from one manufacturer. Two lots of detonators from one manufacturer may be mixed in an S&A Device lot when one detonator lot is completely depleted and a second lot is required to complete the S&A Device lot."

PAGE 10

Paragraph 4.3.2.1, change as follows:

Change Minor AQL to 2.50 Percent

Add: Minor 205 - "205. Protective finish improper (if applicable) - Visual - 01009"

PAGE 13

** Para 4.3.2.5 - Add:

104. Staking between Spur Gear and Pinion improper - Torque Wrench 05005

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Paragraph 4.3.2.26, Delete: Major 104

PAGE 29

Add: Minor 210 - "210. Protective finish improper - Visual - 26027"

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Paragraph 4.3.2.28, Add: Minor 204 - "204. Protective finish improper - Visual - 28015"

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Paragraph 4.3.2.29, change as follows:
Add: Minor 203 - "203. Protective finish improper - Visual - 29006"

Change: Minor AQL from "1.00" Percent to "1.50" Percent.

Paragraph 4.3.2.30, change as follows:

Change: Title from "Leaf and Pin Assembly" to "Leaf, gear, and pin assembly."

Add: Major 104 as follows:

"104	Runout of spur gear to shaft	Hardinge chuck Height gage Dial Indicator	30004"

Add: Major 105 as follows:

"105. Staking insecure - Visual/Manual - 30005"

Change: Major AQL from "0.65" Percent to "1.00" Percent

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Paragraph 4.3.2.31, change as follows:

Change: Minor AQL from "1.00" Percent to "1.50" Percent

Add: Minor 203 as follows: "203. Protective finish improper (if applicable) - Visual - 31009"

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Paragraph 4.3.2.33, Delete: Minor 201

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Paragraph 4.3.2.35 add major defect 108 as follows:
"108 width of spring tab"

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Paragraph 4.3.2.36, change as follows:

Detail drawing number

Was: "9220951"

Now: "9318160"

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Paragraph 4.3.2.37, change as follows:

Detail drawing number

Was: "9220951"

Now: "9318160"

Paragraph 4.3.2.37, add Major Defect 101 as follows:

"Major 101.	AQL 0.40 percent Diameter of wire	Gage	37004"

Paragraph 4.3.2.39 change to read as follows:

"4.3.2.39 Base, Safety and Arming Device (see dwg 9318161 covering a detail of dwg 9318160)

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Categories	Defects	Meth. of Insp.	Code No.
Critical	None defined		
Major	AQL 0.65 percent		
101.	Distance between support mounting surfaces	Vernier Calipers	39013
102.	Location of No 3 electrical contact threads	Fixture	39014
103.	Location of support plate mounting holes	Fixture	39015
Minor	AQL 0.65 percent		
201	Evidence of poor workmanship (see 3.13)	Visual	39016"

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Paragraph 4.3.2.40, change as follows:

Detail drawing number

Was: "9220951"

Now: "9318159"

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Paragraph 4.3.2.41, Add Major Defect 105 as follows:

"105	Detent Latch Stop not fully seated on base"	Visual	41005

Paragraph 4.3.2.41, Delete this para in its entirety.

Paragraph 4.3.2.42, change to read as follows:

"4.3.2.42 contact, electrical, stud (see dwg 9318166 covering a detail of dwg 9318164)

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Categories	Defects	Meth of Insp	Code No
Critical	None Defined		
Major	AQL 0.40 percent		
101.	Diameter of small pin	Gage	42005
102.	Diameter of surface adjacent to pin	Gage	42006
Minor	AQL 0.65 percent		
201.	Protective finish	Visual	42004"

Paragraph 4.3.2.43 change to read as follows:

"4.3.2.43 Base, terminal board (see dwg 9318167 covering a detail of dwg 9318164)

Categories	Defects	Meth of Insp	Code No
Major	AQL 0.65 percent		
101.	Diameter of contact electrical stud counterbore	Gage	43015
102.	Length of contact electrical stud counterbore	Gage	43016
103.	Diameter of No 1 contact hole	Gage	43017
104.	Length of No 1 contact hole	Gage	43018
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.13)	Visual	43019"

Paragraph 4.3.2.43

Add major defects 105 and 106 as follows:

"105. Diameter of shoulder pin cavity"

"106. Depth of shoulder pin cavity"

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Paragraph 4.3.2.4.4, change to read as follows:

"4.3.2.44 Terminal Board and Contact assembly (see dwg 9318164 covering a detail of 9318163)

Categories	Defects	Meth of Insp	Code No
Critical	None defined		
Major	AQL 0.40 percent		
101.	Contacts loose or damaged	Push Test	44003
Minor	None defined"		

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Paragraph 4.3.2.45, add Major Defect 101 as follows:

Categories	Defects	Meth of Insp	Code No
"Major	AQL 0.40 percent		
101.	Base charge weight incorrect	Scale	45010"

Paragraph 4.3.2.46, change to read as follows:

"4.3.2.46 Terminal Board assembly, Electrical-Base (see dwg 9318163 covering a detail of 9318160)

Categories	Defects	Meth of Insp	Code No
Critical	None defined		
Major	AQL 0.40 percent		
101.	Parts loose or damaged	Visual/Manual	46005
102.	Wires touching	Visual	46006
Minor	None defined"		

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Paragraph 4.3.2.47, change to read as follows:

"4.3.2.47 Contact, Electrical, No 3 (see dwg 9318162 covering a detail of dwg 9318160)"

Categories	Defects	Meth of Insp	Code No
Critical	None defined		
Major	AQL 0.40 percent		
101.	Pitch diameter of external thread	Gage	47007
102.	Pitch diameter of internal thread	Gage	47008
103.	Length of internal thread, min.	Gage	47009
104.	Outside diameter	Gage	47010
Minor	AQL 1.00 percent		
201.	Evidence of poor workmanship	Visual	47011
202.	Protective finish	Visual	47012"

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Paragraph 4.3.2.48, change as follows:

Detail drawing number

Was: "9220959"

Now: "9318159"

Paragraph 4.3.2.49, change as follows:

Detail drawing number

Was: "9220959"

Now: "9318159"

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Paragraph 4.3.2.51, change as follows:

S & A Mechanism Assembly drawing number

Was: "9220959"

Now: "9318159"

Add Major Defect 103 as follows:

"103 Support plates not seated on base Visual 51003"

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Paragraph 4.3.2.51, delete Major Defect 102.

Paragraph 4.3.2.52, change to read as follows:

"4.3.2.52 Cover, Access (see dwg 9318169 covering a detail of 9318168)

Categories	Defects	Meth of Insp	Code No
Critical	None defined		
Major	AQL 0.65 percent		
101.	Pitch diameter of ground wire thread	Gage	52011
102.	Length of ground wire thread, min	Gage	52012
103.	Pitch diameter of center boss thread	Gage	52013
104.	Length of center boss thread, min	Gage	52014
** 105.	Minor dia. of ground wire thread, min	Gage	52016
** 106.	Minor dia. of center boss thread	Gage	52017

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	Categories	Defects	Meth of Insp	Code No
**	Major	AQL 100%		
**	107	Minor dia. of ground wire thread, max.	Gage	52018
	Minor	AQL 0.65 percent		
	201.	Protective finish	Visual	52015"

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Paragraph 4.3.2.53, change to read as follows:

"4.3.2.53, Access cover assembly (see dwg 9318168 covering a detail of dwg 9318160)

Categories	Defects	Meth of Insp	Code No
Critical	None defined		
Major	100 percent		
101.	Soldering inadequate or incomplete	Test (4.4.15.4)	53007"

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Paragraph 4.3.2.55, Change as follows:

Was: "AQL 0.65 Percent"

Now: "AQL 1.00 Percent"

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Paragraph 4.3.2.55, Add the folowing Major Defects:

"107	Large inside diameter	Gage	55011
108	Depth of large inside diameter	Gage	55012"

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Paragraph 4.3.2.55; change as follows:

Change Minor AQL to "2.50" Percent

Add: Minor 205 - "205. Protective finish improper (if applicable) Visual - 55011"

Paragraph 4.3.2.56, change as follows:

Change Major AQL to "0.40" Percent

Delete: Major defects 101 and 103

Change Major 102 as follows:

Was:

"Evidence of voids, holes or incomplete seal between window and housing.....Visual 56002"

Now:

"Evidence of wrinkles voids, holes, poor adhesion, or incomplete seal between window and housing....Visual 56002"

Paragraph 4.3.2.58, change as follows:

Change title to read as follows: "Housing, Lead Assembly (see dwg 9241447)"

Change Major 101 as follows:

Was:

"101 Lead charge missing or improperly assembled - green showing - Visual 58001"

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Now:

"101 Lead charge missing or improperly assembled - no blue showing - Visual 58001"

Paragraph 4.3.2.59, change title to read as follows: "Lead, Explosive, PA 501 (see dwg 9220893 covering a detail of dwg 9241447)"

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Paragraph 4.3.2.60, change as follows:

Delete: Critical Defect 3
Add: Special Defect as follows:

"Special			
a.	Shorting disc missing, loose or does not make contact on two pins	Visual/ Manual	60004"

Add: Major Defect 104 as follows:

"104	Detent Latch Stop positioned against Detent Latch improperly	Visual	60008"

Paragraph 4.3.2.61, delete this para in its entirety.

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Paragraph 4.3.2.62, delete this para in its entirety.

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Paragraph 4.3.2.63, delete this para in its entirety.

Paragraph 4.3.2.64, delete this para in its entirety.

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Paragraph 4.3.2.65, delete the present paragraph and substitute the following: "4.3.2.65 Pin, Latch (see dwg 9297946 covering a detail of dwg 9220902).

Categories	Defects	Meth of Insp	Code No
Critical	None defined		
Major:	AQL 0.65 Percent		
101	Diameter of knurl	Gage, Ring go/no go	65001
102	Small outside diameter	Gage, Ring go/no go	65002
103	Surface finish, knurl to chamfered end	Profilometer	65003
Minor:	AQL 1.00 Percent		
201	Evidence of poor workmanship (see 3.13)	Visual	65004"

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Paragraph 4.3.2.67, delete the paragraph in its entirety.

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Paragraph 4.3.2.68, change as follows:

Change Major 102 as follows:

Was:

"102. Parts improperly assembled - Visual-68002"

Now:

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"102. True position of Spacers to Datum A - Fixture-68005"

Delete: Minor 202

Change Minor AQL to "0.65" Percent

** Paragraph 4.3.2.69 - Add:

Minor: AQL 0.65 Percent

201. Height of spacer sleeve rivet heads ----- Gage 69003

Paragraph 4.3.2.69, change as follows:

Change Major 101 as follows:

Was:

"101 Parts improperly assembled - Visual - 69001"

Now:

"101 True position of Spacers to Datum A & B - Fixture - 69004"

Paragraph 4.3.2.70, delete this para in its entirety.

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Paragraph 4.3.2.72, change as follows:

Delete: Minor 201

Change Minor AQL to "0.65" Percent

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Paragraph 4.3.2.75, delete this para in its entirety.

Paragraph 4.3.2.76, Delete this paragraph in its entirety and replace it with the following:

"4.3.2.76 Disc, Electrical-Shorting (see dwg 9326995 covering a detail of dwg 9220961).

Categories	Defects	Meth of Insp	Code No
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Hole diameter	Gage	76004
102.	Outside diameter	Gage	76005
103.	True position of tabs and notches with hole diameter	Gage	76006
104.	Formed angle of tabs	Gage	76007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.13)	Visual	76008"

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Paragraph 4.3.2.77, delete components associated with drawings 9220987, 9233191, 9220984, 9220985. Add the following:

Drawing No	Nomenclature
"9318170	Grommet, Rubber"

Paragraph 4.3.2.77, Add the following:

Drawing No	Nomenclature
9327955	Pin, Shouldered"

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Add: Paragraph 4.3.2.78 as follows:

"4.3.2.78 Spacer, Pallet (See dwg 9220906 covering a detail of 9317905)

Categories	Defects	Meth of Insp	Code No
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Width of Spacer, Max	Gage	78001
Minor:	None defined"		

Add paragraph 4.3.2.79 as follows:

"4.3.2.79 Stop, detent latch (see dwg 9318155 covering a detail of 9220951)

Categories	Defects	Meth of Insp	Code No
Major:	AQL 1.50 percent		
101.	Length from center of hole to base	Gage	79001
102.	Length from center of hole to outside end of support arm	Gage	79002
103.	Length from base to top of support arm	Gage	79003
104.	Thickness of latch stop	Gage	79004
Minor:	AQL 2.50 percent		
201	Diameter of hole	Gage,Ring Go/No-Go	79005"

Table I, change as follows:

Applicable drawing number

Was: "9220959"
Now: "9318159"

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Table I, change as follows:

Applicable drawing number

Was: "9220980"
Now: "9318168"

Table I, Change to the following:

TEST OR EXAMINATION	TEST METHOD PARAGRAPH	INSP EQUIP REQUIRED
Dielectric Strength		
Rotor Assembly	4.4.6.1	
S&A Base Assembly	4.4.6.2	MEGOHMMETER

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Table I - continued, last line, Delete: "Shorting Clip" and Substitute: "Shorting Disc".

Table I, Add the following Test Category"

T or E	Def Code No	Reg	T.M. Par	No of S	App Draw	I.E. Req
"Front Support & Spacer Assembly						
Spacer Push Test	69004	9241388	4.4.22	10	9241388	Static Load Fixture
Cross Section Exam	69005		4.4.23	10	9241388	Visual"

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Table II, change as follows:

Applicable drawing number

Was: "9220959"

Now: "9318159"

Table II, Under Piston Actuator at -40°F, add the following:

T or E	Def Code No	Reg	T.M. Par	No of S	App Draw	I.E. Req
"Pin to Case Resistance After Firing	45011	9220954	4.4.2.3.1.2	--	9220954	-"

Table II, Under Piston Actuator at +145°F, add the following:

T or E	Def Code No	Reg	T.M. Par	No of S	App Draw	I.E. Req
"Pin to Case Resistance After Firing	45012	9220954	4.4.2.3.2	--	9220954	-"

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Table II, change as follows:

Applicable drawing numbers

Was: "9220959"

Now: "9318159"

Was: "9220980"

Now: "9318168"

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Table II, Change to the following:

TEST OR EXAMINATION	TEST METHOD PARAGRAPH	INSP EQUIP REQUIRED
Dielectric Strength		
Rotor Assembly	4.4.6.1	
S&A Base Assembly	4.4.6.2	MEGOHMETER

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Table II - continued, line 3, Delete: "Shorting Clip" and Substitute: "Shorting Disc"

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Table II, Add the following category:

Test or Examination	Def Code No	Req	Test Meth Para	No of Samples	Appl dwg	Insp Equip Req
"Front Support & Spacer Assembly						
Spacer Push Test	69004	9241388	4.4.22	(c)	9241388	Static Load Fixture
Cross Section Exam	69005		4.4.23	(c)	9241388	Visual

(c) MIL-STD-105, Special Inspection Level S-4, Single Sampling Plan, AQL = .65 Percent."

Table II, add the following:

T or E	Def Code No	Req	T.M. Par	No of S	Appl Draw	I.E. Req
"Piston Actuator Functioning All Fire at -65°F	79001	9220954	4.4.20	20	9220954	Power Supply Current

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T or E	Def Code No	Reg	T.M. Par	No of S	App Draw	I.E. Req
No fire at +160°F	79002	9220954	4.4.21	20	9220954	Power Supply Current"

Table II, add the following examination:

T or E	Def Code No	Reg	T.M. Par	No of S	App Draw	I.E. Req
"Rotor pin must touch top of support slot (armed position)	51002			100%	9318159	Visual"

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Table III, last column, change as follows:

Production lot size was: "1201-3000" - now: "1201-6000"

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Paragraph 4.4.2.1 change to read as follows:

"4.4.2.1 Electrical Resistance of the S & A Mechanism Assembly (9318159)".

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"Unarmed condition - Using an electrical test meter limiting current to one (1) milliampere maximum, connect the meter across the rotor electrical contact (9220914) and the center conductor of the coaxial cable connector (9233186) in the cover assembly."

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Paragraph 4.4.2.1.1, change as follows:

Was: "center conductor of the coaxial cable connector (9233186) in the cover assembly."

Now: "Contact, electrical, No 3 (9318162)"

Paragraph 4.4.2.1.2, delete 1st sentence and change to read as follows:

"Armed condition - With the rotor in the "armed" position and using an electrical test meter limiting the current to one (1) milliampere maximum, connect the meter across the contact electrical, stud ((9318166) and the contact, electrical, No. 3 (9318162).

Paragraph 4.4.2.2.1, change as follows:

Was: "center conductor of the connector (9233186)"

Now: "electrical, contact, NO 3 (9318162)"

Paragraph 4.4.2.2.2, change as follows:

Was: "Center conductors of the connector (9233186)"

Now: "electrical, contact No 3 (9318162)"

Was: "electrical, receptacle, connector plug (9220901-1)"

Now: "contact electrical, stud (9318166)"

Paragraph 4.4.2.3.1.2, change to read as follows:

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"4.4.2.3.1.2 Functioning.-After testing for pre-fire resistance and with the piston actuator mounted in the required test fixture, the units shall be functioned according to the following method or equivalent. With the required load (10 pounds) applied to the piston end of the actuator and the proper series resistance (10 ohms) in the firing circuit, the required voltage (18 volts) shall be applied for the specified pulse duration (4 milliseconds). The force gage shall be capable of measuring the initial load as well as the maximum peak reaction (120 pounds) and the test equipment shall be capable of measuring the minimum required piston travel (0.12 inch). The piston actuator shall be functioned and piston motion, post fire resistance, post fire pin to case resistance, and functioning time measured. The post fire resistance shall be measured within 25 milliseconds after application of current using approved test equipment. The post fire pin to case resistance shall be measured from each lead wire to the piston actuator housing. Functioning time shall be measured to the nearest one tenth of a millisecond. Functioning time is defined as the interval between application of the required voltage to completion of the required minimum distance of travel (0.12 inch)."

Paragraph 4.4.2.3.2, add the following:

"g. The post fire pin to case resistance is less than the minimum drawing requirement."

Paragraph 4.4.2.3.4, change as follows:

Was: "connector receptacle, electrical plug (9220901-1)"

Now: "contact, electrical, stud (9318166)"

Paragraph 4.4.2.3.4, line 9, Delete: "shorting clip" and Substitute: "shorting disc"

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Paragraph 4.4.3, change as follows:

S&A mechanism assembly drawing

Was: "9220959"
Now: "9318159"

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Paragraph 4.4.4.1, change as follows:

S&A mechanism assembly drawing

Was: "9220959"
Now: "9318159"

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Paragraph 4.4.6 Change to read as follows:

"4.4.6 Dielectric Strength"

Add paragraph 4.4.6.1 to read as follows:

"4.4.6.1 Rotor Assembly Dielectric Strength - The specified electrical potential shall be applied between the points to be tested using an AC power supply (0-300 Volts RMS) and an AC microammeter. Breakdown shall be defined as a leakage current in excess of two (2) microamperes for the rotor assembly test under the requirement conditions specified in 3.9.1".

Add paragraph 4.4.6.2 to read as follows:

"4.4.6.2 S&A Base Assembly Dielectric Strength - A minimum of 100 megohms shall be required using a megohmmeter with a minimum of 500 ± 50 volts D.C. under the requirement conditions specified in 3.9.2".

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Paragraph 4.4.11, lines 8 and 10, Delete: "shorting clip" and Substitute: "shorting disc" where it appears.

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Paragraph 4.4.19.1 Change as follows:

"4.4.19.1 Arming Test.-After environmental conditioning and testing, selected S&A Devices shall be subjected to an arming test. The device shall be mounted so that the direction of acceleration simulates missile flight, i.e., top of lead charge directed towards the center of rotation, and the missile battery harness slot located at top vertical center using test equipment as specified. The piston actuator resistance shall be measured on all units which have not been previously tested as specified in 4.4.19.2, and the actuator fired utilizing the firing circuit specified in 4.4.2.3.1.2. The S&A Device shall be accelerated to a level corresponding to 300g's. The centrifuge shall then be stopped causing the rotor to arm. If any unit fails the piston actuator resistance tests or fails to arm as determined by the electrical resistance test of the detonator ciruit, the lot shall be rejected. The detonator resistance shall be measured by connecting the limited current meter between the two insulated terminals. Using the current limited meter the open circuit resistance between the electrical contact, No 3 (9318162) and the electrical receptacle connector plug (9220901-2) shall be checked in the armed position as specified in 4.4.2.2.1. Any unit with a resistance reading less than the minimum requirement of Para 3.18 shall be classed defective. Each S&A Device shall be shook and rotated 360° (end over end) to check for loose parts. Any units exhibiting evidence of loose parts shall be classed defective. The device shall then be subjected to electrical functioning."

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Paragraph 4.4.19.2 Change to read as follows:

"4.4.19.2 Bias Mass and Rotor Delay Time Test.-After environmental conditioning and testing, selected S&A Devices shall be subjected to a bias mass time and rotor delay time test. The S&A Device to be tested shall have its housing assembly (dwg 9241447) removed. The device shall then be placed in a suitable test fixture and conditioned to the appropriate test temperature ($\pm 5^{o}F$). The device shall then be mounted in the centrifuge so that the access cover end is directed away from the center of rotation, and the missile battery harness slot on the access cover is located at the top vertical center. The centrifuge shall be run up to an RPM corresponding to an acceleration level of 300g's at the center of gravity of the bias mass. With the acceleration level stabilized, the bias mass retaining lever shall be mechanically tripped, permitting the bias mass to move and release the rotor for arming. The time interval for bias mass functioning shall be read on the counter to the nearest tenth of a millisecond. If any unit fails to meet the bias time of 3.6 or if the rotor fails to turn to the armed position, the S&A Device shall be classed defective. After recording bias mass functioning time, the centrifuge shall be stopped. With the centrifuge at rest, the rotor shall be reset, the rotor released and the rotor delay time measured and recorded. At least once each working shift, the counter readouts shall be correlated by comparison with the pulse width observed on an oscilloscope. Any device failing to arm within the required time limits specified in 3.7.2 shall be classed defective. At the conclusion of the rotor delay time, the S&A Device shall once again be orientated in a suitable test fixture with the missile battery harness slot located at top vertical center. The piston actuator resistance shall be measured as specified in 4.4.2.3.4 and the actuator fired utilizing the firing circuit specified in 4.4.2.3.1.2. If any unit fails the piston actuator resistance check, the lot shall be rejected. The S&A Device shall be visually inspected after firing the actuator, and if there is any evidence of loose, broken, or missing parts, the lot shall be rejected.

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The housing assembly shall then be reassembled to the S&A Device so it can be tested further (to arming test according to 4.19.1 and electrical functioning according to 4.4.9). (See Tables I and II). The lot shall be rejected if any individual rotor delay time fails the minimum requirement of if the average delay time plus or minus two (2) sigma at any one temperature fails the requirement.

WARNING: These tests are dangerous and should be performed behind a suitable barricade and in a space provided with electrostatic shielding.

NOTE: The following symbols are defined.

$$\bar{X} = \frac{\Sigma \text{Delay Times}}{n}$$

Where: $\bar{x}$ = Average delay time at one temperature
Σ = Sum
n = Number of delay times at one temperature

$$\text{Sigma} = \sqrt{\frac{\Sigma(x - x)\ 2}{n - 1}}$$

Where: Sigma = Sample Standard Deviation
Σ = Sum
x = Individual delay time from one temperature
n = Number of delay times at one temperature"

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Add Paragraph 4.4.20 as follows:

"4.4.20 Piston Actuator All Fire Test.-Twenty (20) samples shall be selected from each lot and conditioned at -65°F for a minimum of two (2) hours. Each temperature conditioned piston actuator shall be subjected to the required all fire current. The lot shall be rejected if any unit fails to function."

Add paragraph 4.4.21 as follows:

"4.4.21 Piston Actuator No Fire Test.-Twenty (20) samples shall be selected from each lot and conditioned at +160°F for a minimum of (2) hours. Each temperature conditioned piston actuator shall be subjected to the required no fire current. The lot shall be rejected if any unit functions."

Add Paragraph 4.4.22 as follows:

"4.4.22 Spacer Push Test.-The Front Support and Spacer Assembly (dwg 9241388) shall be assembled to the test fixture as specified. A force of one hundred and twenty (120) pounds shall be applied to the Spacer, Sleeve-Latch in the direction shown. Any spacer which moves in the direction of the applied force and takes a permanent set shall be classed defective and the lot shall be rejected. (Destructive test sample)".

Results of the push test shall be submitted to: Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-M. (See Par 6.4).

Add Paragraph 4.4.23 as follows:

"4.4.23 Cross Sectioning of Front Support and Spacer Assembly Samples.-The same samples used in the Spacer Push Test (Par 4.4.22) shall be cross-sectioned through the Detent Latch Spacer (9220970) in the same plane as the direction of the static load shown on dwg 9241388. One section from each sample shall be metalurgically prepared to allow visual

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examination of anomolies in the rolled over area of the spacer. The examination will be made under 30X magnification and photographs shall be taken and submitted to: Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-M. (See Para 6.4)"

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Add Paragraph 6.4 as follows:

"6.4 Submission of Inspection Data.-Deliverable data and photographs required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.22	Push test results	DI-R-1724
4.4.23	30X Photographs	DI-R-1724"

**NOTE: The margins of this amendment are marked with two asterisks to indicate where changes from the previous revisions were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

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Preparing Activity
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